

SUBJECT GUIDE FOR:	2026
COURSE CODE:	APT470S
NQF LEVEL:	7

APPLICATIONS DEVELOPMENT THEORY 4

ASSESSMENT RUBRIC

AI-Driven Model Engineering: A Systematic Literature Review of Intelligent Model Generation, Transformation, Validation, and Software Synthesis

Assessment guidance: The project should demonstrate a rigorous, transparent and reproducible systematic literature review. Marks are awarded according to the quality of the evidence, critical analysis, methodological execution, scholarly contribution, and professional presentation. Merely describing studies without synthesis, evaluation, or traceability will not attract high marks.

Criterion	Marks	Excellent (75-100%)	Competent (60-74%)	Developing (50-59%)	Insufficient (0-49%)
1. Research Significance	2	Provides a compelling, well-contextualized research significance of AI-DME; defines key concepts precisely and supports claims with high-quality recent literature.	Provides a clear significance; defines most concepts correctly; significance is explained but may lack depth or strong synthesis.	Significance is generally stated but broad, weakly justified or descriptive; key concepts are partially defined; limited connection to research.	Significance is unclear, unsupported or unrelated to AI-DME; concepts are confused or absent; little evidence of research significance.
2. Research Objectives	3	Objectives and research questions are fully aligned, logically structured, feasible and explicitly cover the AI-DME lifecycle; scope and boundaries are precise and defensible.	Good alignment and coverage with minor gaps, duplication or wording issues; scope is mostly clear.	Partial alignment; objectives or RQs are too broad, repetitive or difficult to answer; scope is vague.	Aim, objectives and RQs are inconsistent, incomplete or not researchable; scope is missing or inappropriate.
3. Review Protocol and Methodological Rigour	5	Presents a complete, reproducible SLR protocol using recognised SE review guidance and PRISMA; clearly specifies databases, period, search procedures, reviewer roles, deduplication, screening, snowballing and protocol deviations.	Sound methodology with most protocol elements specified; minor omissions do not substantially affect reproducibility.	Basic review process is described but lacks important procedural details, reviewer independence or reproducibility.	Method is ad hoc, non-systematic or inadequately described; no credible protocol or PRISMA process.
4. Search Strategy, Study Selection and Eligibility	5	Develops comprehensive, database-adapted search strings with justified keywords and synonyms; applies explicit inclusion/exclusion criteria consistently; documents selection using complete PRISMA flow and transparent exclusion reasons.	Search strategy and eligibility criteria are appropriate with small omissions; selection process is adequately documented.	Search terms or criteria are limited, inconsistently applied or weakly justified; selection documentation is incomplete.	Search is narrow, biased or non-reproducible; criteria are missing or selection decisions are unsupported.
5. Quality Assessment and Risk of Bias	10	Applies a well-justified quality appraisal instrument addressing research clarity, AI/model transparency, data quality, baselines, semantic validity, reproducibility and	Quality assessment is relevant and mostly complete; scoring and use of results are generally appropriate.	Checklist is superficial, inconsistently applied or disconnected from the synthesis.	No credible quality assessment or risk-of-bias analysis is performed.

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		threats to validity; uses scores appropriately in synthesis and evidence weighting.			
6. Data Extraction, Coding and Traceability	10	Uses a comprehensive, pilot-tested extraction framework; coding decisions are explicit and reproducible; data accurately capture AI techniques, modelling tasks, artefacts, datasets, validation, maturity, costs and limitations; full traceability to primary studies is maintained.	Extraction and coding are systematic and mostly complete; minor gaps in traceability or consistency.	Extraction is incomplete or inconsistently coded; limited coverage of important variables; traceability is weak.	Data extraction is unsystematic, inaccurate or largely absent; findings cannot be traced to studies.
7. Evidence Synthesis and Critical Analysis	15	Integrates descriptive, thematic and comparative synthesis with strong critical reasoning; distinguishes evidence strength from claims; compares approaches, baselines, datasets, validation mechanisms and industrial readiness; avoids unsupported generalisations; quantitative synthesis is used only when appropriate.	Provides coherent synthesis and useful comparison; some critical depth or integration may be missing.	Mainly summarises individual studies; limited comparison, synthesis or critical interpretation.	Findings are fragmented, inaccurate or purely descriptive; no meaningful synthesis or evidence-based conclusions.
8. Taxonomy, Maturity Model, Evaluation Hierarchy and Conceptual Framework	15	Produces original, logically derived and well-validated scholarly artefacts; dimensions are mutually clear, comprehensive and grounded in evidence; framework shows variables/constructs and relationships diagrammatically; artefacts demonstrate clear theoretical and practical value.	Artefacts are relevant and mostly evidence-based, though some categories, relationships or validation need refinement.	Artefacts are preliminary, overlapping, weakly justified or largely adapted without sufficient synthesis.	Required artefacts are missing, incoherent, unsupported or disconnected from the review findings.
9. Reference Architecture, Research Agenda and Scholarly Contribution	5	Proposes a clear verifier-guided, standards-compliant, traceable and human-accountable architecture; identifies prioritised, actionable research gaps and contributions grounded directly in evidence.	Architecture and agenda are relevant and generally supported; prioritisation or novelty could be stronger.	Architecture or agenda is generic, weakly linked to evidence or insufficiently actionable.	Little or no credible architecture, research agenda or scholarly contribution is presented.
10. Academic Writing, Referencing and Visuals	5	Exceptionally clear, coherent and concise academic writing; accurate terminology; high-quality diagrams/tables; consistent referencing style; complete reference list; polished formatting; ethical use of AI and transparent acknowledgment where required.	Professional writing and presentation with minor language, citation or formatting errors.	Meaning is generally understandable but writing, structure, visuals or referencing contain recurring weaknesses.	Poorly written, disorganised or inadequately referenced; visuals are unclear or absent; serious formatting or academic-integrity concerns.
11. Quality of Academic Paper used and submitted + Data Extraction (Excel)	10	Adequate	Very Good	Good	Insufficient
12. Oral Professional Presentation & Slides	10	Excellent	Very Good	Good	Fair
13. Others (participation, my books, etc.)	5	Excellent	Very Good	Good	Fair
TOTAL	100				

Overall Performance Bands:

Final Mark	Classification	General Interpretation
75-100	Excellent	Methodologically rigorous, critically synthesised and capable of producing a strong scholarly contribution.
60-74	Competent	Sound and generally well-executed, but requires targeted strengthening in analysis, methodological transparency or contribution.
50-59	Developing	Meets minimum expectations but contains substantive weaknesses in rigour, synthesis, traceability or scholarly contribution.
0-49	Insufficient	Does not meet the required standard for a systematic literature review research project.